SAFETY AND RISK ANALYSIS

# Final Project – In the News

A construction safety case study on temporary structural stability, warning signs, and OSHA enforcement

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# Boise Airport Hangar Collapse

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**JAN 31, 2024 · BOISE, IDAHO**

Partially built aircraft hangar collapsed during steel erection at the Jackson Jet Center expansion project.

## 3 workers killed · 9 injured

OSHA later found visible warning signs and inadequate temporary support during steel erection.

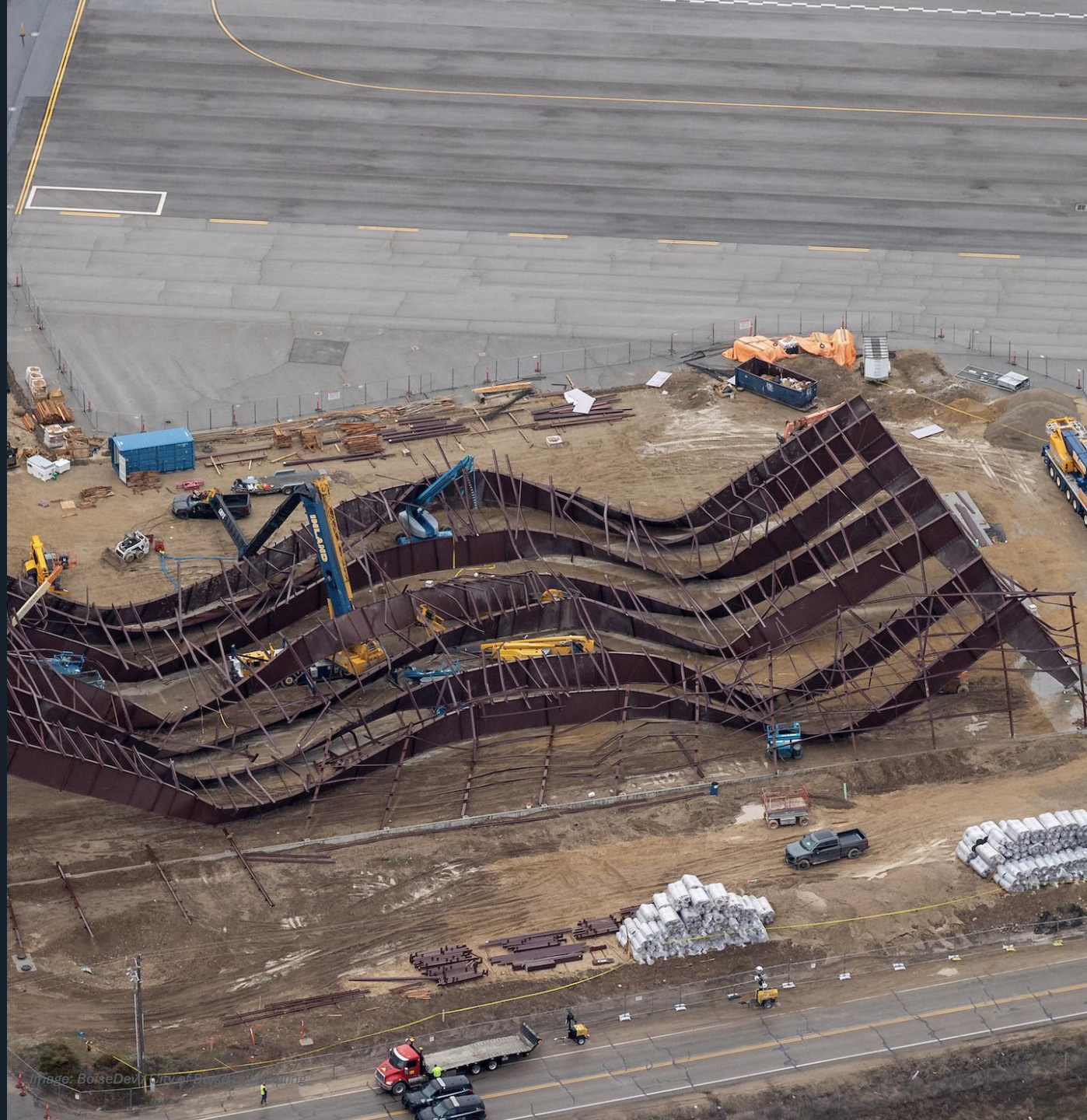



Image: BoiseDev / City of Boise / Idaho Statesman

# Workers died because warning signs were ignored

OSHA described the collapse as preventable and tied it to visible instability and unsafe construction practices.

**3**
**workers killed**
Fatal injuries at the construction site

**9**
**workers injured**
City/AP reporting; OSHA describes at least 8 injured workers

**$6.2M**
**project value**
39,000-sq-ft hangar permitted for Jackson Jet Center

**1 + 3**
**OSHA violations**
1 willful + 3 serious citations for Big D Builders

*Image: KTVB/AP archive image of collapsed steel/wreckage*

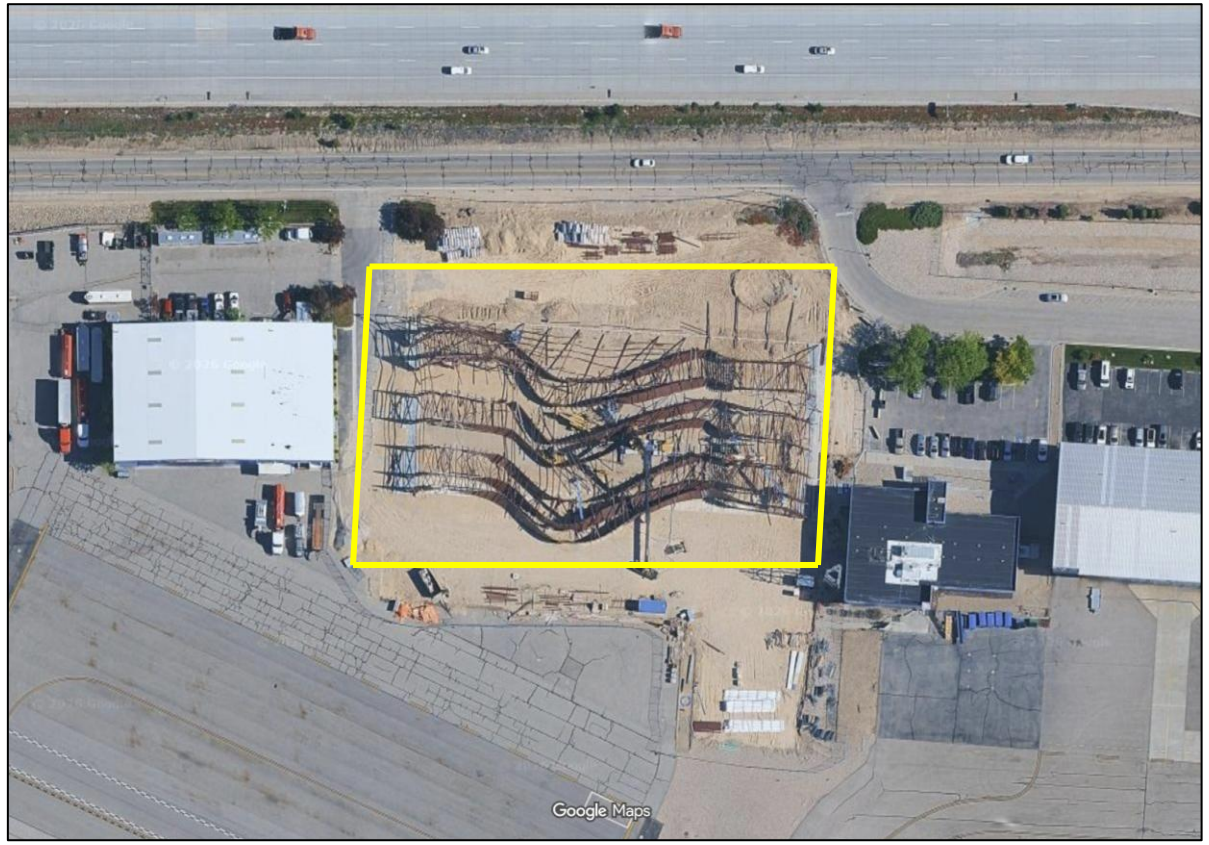
### News angle

- Visible curved, bent, and wavy I-beams were reported before collapse.
- Big D continued adding 150-foot-long bays despite instability concerns.
- The case became a national example of steel-erection hazard control.

**Corrected penalty wording: OSHA proposed $198,586 for Big D Builders; Inland Crane was separately cited for $10,163.**

# Jackson Jet Center hangar site, Boise Airport

The collapse happened inside Boise Airport property, near W. Rickenbacker St., Luke St., and W. Wright St.




*Location image: Google Maps satellite imagery, accessed April 27, 2026.*

### Project

Jackson Jet Center hangar expansion: 39,000 sq ft steel-framed aircraft maintenance hangar.

### Contractor

Big D Builders Inc. was responsible for the construction work; Inland Crane was also cited by OSHA.

### Exposure

Multiple workers were on or near the partially erected steel frame. The frame depended on temporary bracing and guy wires until it became self-supporting.

### Key background insight

A building under construction is a temporary structure. Every bay, bolt, and brace changes the load path during steel erection.

# What happened, step by step

OSHA's findings connect the collapse to inadequate temporary support, visible instability, and continued work.

**1** → **2** → **3** → **4** → **5**

**Steel bays added**

Frame erected bay by bay

**Insufficient bracing**

Not enough tensioned guy wires

**Warning signs visible**

Curved/wavy beams and unbalanced columns

**Work continued**

Loose bolts and straps used instead of proper bracing

**Global collapse**

Steel frame failed; 3 killed and 9 injured

## OSHA's core finding

Big D Builders began erecting the hangar without sufficient bracing or tensioned guy wires; warning signs included bent/wavy beams, unbalanced columns, and snapped wire rope cables.

*Aerial image of collapsed hangar: BoiseDev / City of Boise public filing.*

# Six warning signs and control failures

These were present before collapse and should have triggered stop-work and engineering review.

### Inadequate bracing

Insufficient bracing or tensioned guy wires for the erection sequence.

### Visible deformation

Bent, curved, and wavy structural I-beams showed instability.

### Unbalanced columns

Columns were not straight and did not indicate a stable load path.

### Snapped cables

Snapped wire rope cables were a clear warning sign.

### Loose bolts / straps

Critical bolts were left loose; straps were used instead of proper bracing.

### Training + site conditions

Workers were not properly trained; cranes operated in mud/standing water.

**Risk takeaway: the collapse was not “sudden” from a safety perspective. The structure showed warning signs before the final failure.**

# The fallout went beyond one collapsed structure

The incident created direct project cost, investigation delay, demolition/rebuild work, and legal/insurance exposure.

**$6.2M**

**permitted project value**

Concrete foundation + metal building for Jackson Jet Center

**$198,586**

**Big D proposed penalties**

1 willful + 3 serious OSHA violations

**$10,163**

**Inland Crane penalty**

Separate OSHA citation for collapse-hazard exposure

## Schedule / quality chain

**Collapse** → **OSHA investigation** → **Removal + demolition** → **Engineering revisions** → **Rebuild**

**Quality impact:** Erection QC, supervision, temporary bracing verification, and stop-work authority were not functioning. Human, legal, insurance, and delay costs are not fully public.

# Bowtie view of the steel-erection hazard

Risk controls should reduce both the probability of collapse and the severity if collapse begins.

**Inadequate temporary bracing** →

**Loose / unsecured connections** →

**Warning signs ignored** →

**TOP EVENT**
**Steel frame loses temporary stability**

→ **Worker fatalities and injuries**

→ **Construction halted + demolition**

→ **OSHA penalties + litigation risk**

## Preventive barriers

- Qualified-person erection plan
- Temporary bracing for each sequence step
- Bolt/guy-wire checklist before each lift
- Stop-work authority for all workers

## Mitigating barriers

- Exclusion zones below active steel erection
- Immediate evacuation for instability signals
- Emergency response plan and access routes
- Third-party inspection during critical phases

# From violation to prevention

The official investigation identifies what must change on steel-erection sites.

## OSHA citations · July 29, 2024

**Big D Builders Inc.**

1 willful + 3 serious violations. Proposed penalty: $198,586.

**Inland Crane Inc.**

1 serious violation. Proposed penalty: $10,163 for collapse-hazard exposure tied to stability failures.

**Important note**

Penalties were proposed; employers may comply, request an informal conference, or contest findings.

## Corrective actions to emphasize

- Written steel erection plan before work begins.
- Temporary bracing and guy wires reviewed for each erection stage.
- Do not continue work when beams bend, columns lean, or cables snap.
- Train workers and supervisors to recognize instability and stop work.
- Keep cranes and equipment out of mud or standing water where overturning risk increases.

**OSHA's message: this was preventable because the warning signs existed before the collapse.**

# What this means for construction engineering

Four takeaways that apply to any project involving temporary structures or phased erection.

## 1 Temporary conditions are real structures

During steel erection, the incomplete frame has its own load path and failure modes. It must be engineered as carefully as the finished building.

## 2 Warning signs require action

Leaning columns, bent beams, snapped cables, unusual sounds, or loose bolts are stop-work conditions — not problems to work around.

## 3 Quality control must be auditable

Daily checklists for bolts, bracing, guy wires, and sequence changes create the record that prevents improvisation.

## 4 Training must match the current phase

Steel erection hazards are different from general site hazards; workers and supervisors need phase-specific training and authority.

**Bottom line: do not wait for collapse to prove instability. Temporary deflection, loose connections, and unusual movement are enough to stop work.**

# Sources and image credits

1. U.S. Department of Labor / OSHA. “Department of Labor finds appalling disregard of safety standards by building contractor in collapse of Boise airport hangar that killed 3 workers.” July 29, 2024.

2. City of Boise. “UPDATE: Hangar Collapse Under Investigation.” February 1, 2024.

3. Associated Press. “3 people killed and 9 injured in hangar collapse on grounds of Boise, Idaho, airport.” February 1, 2024.

4. Associated Press. “Idaho coroner releases names of the 3 men who were killed when a Boise aircraft hangar collapsed.” February 2, 2024.

5. BoiseDev. “Timeline set for removal of collapsed Boise hangar, rebuilding planned.” June 18, 2024.

6. OSHA 29 CFR 1926 Subpart R — Steel Erection; cited as the safety framework for steel-erection hazards.

7. Image credits: BoiseDev / City of Boise public filing; Google Maps satellite imagery; KTVB / AP archive image.

# Thank You